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How Costly Are Cultural Biases? Evidence from FinTech

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1 Big picture

- **Question.** Are cultural biases in financial decision-making costly to the people who hold them, and can a simple robo-advising tool attenuate those biases?
- **Core setting.** The paper studies lenders on *Faircent*, an Indian peer-to-peer lending platform, and compares the loans the *same lenders* choose when they invest manually with the loans chosen after they adopt the platform’s robo-advising tool, AUTO INVEST.
- **Two biases.**
 - **In-group vs. out-group discrimination:** Hindu vs. Muslim lenders and borrowers.
 - **Stereotypical discrimination:** bias against *Shudra* (low-caste) borrowers.
- **Why this setting is unusually clean.**
 - Borrowers are screened by the platform before entering the pool.
 - The platform sets loan amount, maturity, and interest rate; lenders mainly choose *which* borrower to fund and *how much*, not the price.
 - AUTO INVEST matches borrowers roughly by order of arrival within broad risk buckets rather than by demographic characteristics.
 - The setting has very little relationship lending: more than 90% of lenders fund borrowers across at least five Indian states.
- **Main empirical result.** When lenders choose manually, they are more likely to fund co-ethnic or socially favored borrowers. After adopting AUTO INVEST, this imbalance shrinks sharply.
- **Main performance result.** The discrimination is costly to the discriminator. Relative to the robo-advised period, unassisted lending is associated with about **8% higher defaults** and up to **7.3 percentage-point lower returns**.
- **Mechanism.** Manually biased lenders appear to *dig deeper into the pool of favored borrowers*: they keep choosing lower-quality co-ethnic or upper-caste-favored borrowers. AUTO INVEST reduces this by removing the demographic tilt from matching. The return gains come mainly from eliminating the *left tail* of very bad loans, not from better diversification.
- **Interpretation.** The evidence fits **biased beliefs / implicit bias** better than taste-based discrimination:
 - discriminated borrowers outperform favored borrowers before robo-advice;
 - lenders rarely override robo-advised matches to the groups they discriminated against when unassisted;
 - the effects are stronger in places where inter-group conflict is more salient.
- **Economic takeaway.** A simple, opt-in FinTech tool that strips demographics out of borrower matching can improve portfolio performance and reduce culturally biased allocation choices at the same time.

2 Data and measurement (key objects)

2.1 Institutional setting: Faircent

- Faircent is a large Indian peer-to-peer lending platform.
- Borrowers apply by submitting a loan request with amount, purpose, credit score, occupation, geographic location, and family/dependent information.
- The platform first screens applicants using an algorithmic credit-viability procedure and then a verification stage. Only screened and verified borrowers enter the lender-visible pool.
- The platform assigns the loan’s key contract terms — especially **interest rate**, **maturity**, and **loan amount**. These terms are not negotiated by lenders.
- To limit money-laundering concerns, each loan must be financed by at least five lenders, and each lender can finance at most 20% of a loan.
- The platform monitors and services loans after origination. Lenders and borrowers do not build local branch-style relationships.

Why this matters. The paper is not asking whether lenders can price risk. The platform has already done a great deal of the screening and pricing. The residual lender decision is mostly an *allocation* decision among screened borrowers, which is exactly where cultural bias can show up.

2.2 Manual investing vs. robo-advising

- **Manual mode.** Lenders browse the borrower pool and observe:
 - coarse risk category (low/medium/high),
 - detailed loan terms,
 - names and surnames,
 - education, occupation, residence, and number of dependents,
 - identity-verification information.
- **Auto Invest.** Introduced in the second half of 2018. Lenders choose:
 - what share of funds to allocate through the tool,
 - how to split those funds across low-, medium-, and high-risk buckets.
- Within each risk bucket, AUTO INVEST matches borrowers approximately by *order of admission to the pool*. The tool does not explicitly optimize on religion, caste, or other borrower demographics.
- Importantly, the paper argues and verifies that AUTO INVEST’s proposed matches are uncorrelated with lenders’ and borrowers’ demographic traits.

Interpretation. AUTO INVEST is not a “better information” device in the sense of revealing new hidden soft information. It is closer to a *demographically blind assignment rule* inside broad risk classes.

2.3 Two empirical samples

- **Religion sample (Hindu vs. Muslim).**
 - 113,283 lender-borrower-loan triads.
 - 2,818 unique lenders.
- **Caste sample (Shudra vs. non-Shudra).**
 - 62,831 lender-borrower-loan triads.
 - Only Hindu lenders and borrowers for whom broad caste/varna can be inferred.

2.4 Summary statistics

- In the religion sample:
 - 13% of borrowers are Muslim.
 - 99% of lenders are Hindu.
 - 45% of loans are issued after AUTO INVEST adoption.
 - The average AUTO INVEST allocation is 59% of platform funds.
 - Average maturity is about 22 months.
 - Average loan amount is about 131,000 rupees.
 - Average interest rate is about 24%.
 - Average delinquency rate is about 29%.
- In the caste sample:
 - 39% of borrowers are identified as Shudra.
 - 25% of lenders are identified as Shudra.
 - Average delinquency is about 28%.

Interpretation. The Muslim-lender sample is small, but the paper still detects economically and statistically meaningful within-lender changes before and after robo-advice.

2.5 How religion and caste are measured

- The authors combine platform records with an Indian marriage-registry dataset covering about 2.48 million individuals.
- **Religion.** Religion is inferred from surname and date of birth. The mapping is very sharp: among Hindus and Muslims, about 96% of surname-date-of-birth pairs map uniquely to one of the two religions.
- **Caste.** Caste is harder to infer. The authors rely on methods from Bhagavatula et al. (2017, 2018) to infer broad varna and, crucially, the *recognizability probability* of caste from observed borrower information.
- This recognizability measure is central to the caste analysis because stereotypical discrimination should be stronger when lenders can more easily infer that a borrower is Shudra.

3 Empirical framework and specifications (all equations + interpretation)

3.1 Religious out-group lending

The baseline religion-allocation regression is

$$\begin{aligned} \text{MuslimBorrower}_{ijt} = & \alpha + \beta \text{AutoInvest}_{jt} + \gamma \text{HinduLender}_j + \delta (\text{HinduLender}_j \times \text{AutoInvest}_{jt}) \\ & + \zeta x_{it} + \eta_j + \eta_t + \varepsilon_{ijt}. \end{aligned} \quad (1)$$

Objects.

- $\text{MuslimBorrower}_{ijt} = 1$ if lender j funds a Muslim borrower i in year t .
- $\text{AutoInvest}_{jt} = 1$ after lender j has adopted AUTO INVEST.
- x_{it} contains platform-assigned loan risk proxies: maturity, loan amount, and interest rate.
- η_j and η_t are lender and year fixed effects.

Interpretation of signs.

- $\gamma < 0$: before robo-advice, Hindu lenders are less likely than Muslim lenders to fund Muslim borrowers — that is, stronger Hindu in-group bias / Muslim out-group bias.
- $\delta > 0$: after robo-advice, Hindu lenders become more likely to fund Muslim borrowers, which means AUTO INVEST reduces religious in-group bias.

3.2 Caste-based stereotypical lending

For caste, the core selection equation is

$$\text{ShudraBorrower}_{ijt} = \alpha + \beta \text{AutoInvest}_{jt} + \zeta x_{it} + \eta_j + \eta_t + \varepsilon_{ijt}. \quad (2)$$

Interpretation.

- $\text{ShudraBorrower}_{ijt} = 1$ if lender j funds a Shudra borrower.
- Here, because the caste sample focuses on Hindu lenders and then studies heterogeneity by caste recognizability and lender type, the key parameter is β .
- $\beta > 0$ means that after robo-advice, lenders allocate more credit to Shudra borrowers, implying that before robo-advice they were discriminating *against* them.

3.3 Loan performance: defaults

For defaults, the paper estimates variations of

$$\begin{aligned} \text{DelinquentLoan}_{ijt} = & \alpha + \gamma \text{MuslimBorrower}_{ij} + \delta (\text{MuslimBorrower}_{ij} \times \text{AutoInvest}_{jt}) \\ & + \theta (\text{HinduBorrower}_{ij} \times \text{AutoInvest}_{jt}) + \zeta x_{it} + \eta_j + \eta_t + \varepsilon_{ijt}. \end{aligned} \quad (3)$$

Interpretation.

- $\gamma < 0$ means discriminated borrowers (Muslims in Hindu lenders' portfolios) default *less* than favored borrowers before robo-advice.
- More negative post-adoption coefficients for favored borrowers imply that robo-advice improves the quality of the *favored-group loans* the most — exactly what we would expect if manual bias makes lenders reach too far down the quality distribution of favored borrowers.

The same logic is used for Shudra vs. non-Shudra borrowers.

3.4 Loan performance: returns

For loan returns, the paper uses the same structure with loan return as the dependent variable:

$$\begin{aligned} \text{LoanReturn}_{ijt} = & \alpha + \gamma \text{MuslimBorrower}_{ij} + \delta (\text{MuslimBorrower}_{ij} \times \text{AutoInvest}_{jt}) \\ & + \theta (\text{HinduBorrower}_{ij} \times \text{AutoInvest}_{jt}) + \zeta x_{it} + \eta_j + \eta_t + \varepsilon_{ijt}. \end{aligned} \quad (4)$$

Interpretation.

- If cultural bias is costly, then discriminated borrowers should earn *higher* returns before robo-advice, because lenders passed them over in favor of lower-quality favored-group borrowers.
- After adoption, returns on favored-group loans should improve the most.

3.5 Quantile regressions

To understand where in the return distribution robo-advice helps, the paper estimates

$$Q_\tau(\text{Returns}_{ij}) = \alpha(\tau) + \beta(\tau) \text{AutoInvest}_j + X'_{ij}\zeta(\tau) + \varepsilon_{ij}. \quad (5)$$

Interpretation.

- If $\beta(\tau)$ is large and positive mainly at low quantiles, robo-advice is removing the *worst* loans.
- If these quantile effects vanish after conditioning on loan risk, then the gain comes from replacing systematically riskier manual choices with safer screened matches.

3.6 Lender-level total returns

At the lender level, the paper defines total return before and after adoption as

$$\text{LenderTotRet}_{i,t} = 100 \times \frac{\sum_j \text{AmountDisbursed}_{ijt} \times \text{LoanReturn}_{jt}}{\sum_j \text{AmountDisbursed}_{ijt}}, \quad (6)$$

and the change in total return as

$$\text{ChangeLenderReturn}_i = \text{LenderTotRet}_{i,POST} - \text{LenderTotRet}_{i,PRE}. \quad (7)$$

Interpretation.

- This is the cleanest “how much money did bias cost the lender?” object.
- Positive values mean the lender earned higher aggregate returns after adopting AUTO INVEST.

4 Identification and why the estimates are credible

4.1 Why the design is persuasive

- **Same lender before/after.** Much of the variation is within lender. This removes many fixed lender traits such as education, financial literacy, and stable screening ability.
- **Only eventual adopters.** The main sample compares not-yet-adopters to already-adopters among lenders who all adopt at some point, reducing concerns that adopters and never-adopters are fundamentally different.
- **Borrowers are pre-screened.** The platform has already done risk screening and pricing. This sharply narrows the scope for a benign interpretation in which manual discrimination simply reflects superior lender screening.
- **Little relationship lending.** Because lenders fund geographically dispersed borrowers and the platform handles monitoring/servicing, the usual in-group monitoring or stigma channels are largely shut down.
- **Common shocks should not produce opposite-signed reallocations.** A borrower-level shock that made Muslim borrowers better or worse should move *all* lenders in the same direction, not induce Hindu and Muslim lenders to move in opposite directions.
- **Heterogeneity aligns with social conflict.** The bias is strongest where Hindu-Muslim riots, BJP political support, or caste-related hate crimes make those cleavages more salient.
- **Performance results have the “right” sign for costly bias.** Discriminated borrowers outperform favored borrowers before robo-advice. This is the opposite of what we would expect if discrimination improved selection.
- **Diversification does not explain the results.** Portfolio-diversification measures barely move after adoption.

4.2 Why the paper leans toward biased beliefs rather than taste-based discrimination

- If lenders had a strong conscious taste for discrimination, they should often override robo-advised matches involving disfavored groups. They rarely do.
- The biggest gains come from removing the worst favored-group loans, which looks more like mistaken beliefs than utility from discrimination.
- Debiasing is strongest where social conflict is salient, which is natural for belief distortions induced by culture and stereotypes.

4.3 Limitations

- Adoption of AUTO INVEST is not randomized.
- The paper cannot observe, at a given moment, whether each loan contribution was made manually or exactly through the robo-advisor when a lender uses both channels.

- Religion and especially caste are inferred, not directly observed by the econometrician.
- The Muslim-lender sample is small.
- Loan-return analyses use a reduced sample because detailed servicing data and loan closure information are required.

5 Tables and figures (with extracted visuals)

Fig. 1. Raw religious in-group bias before and after robo-advice

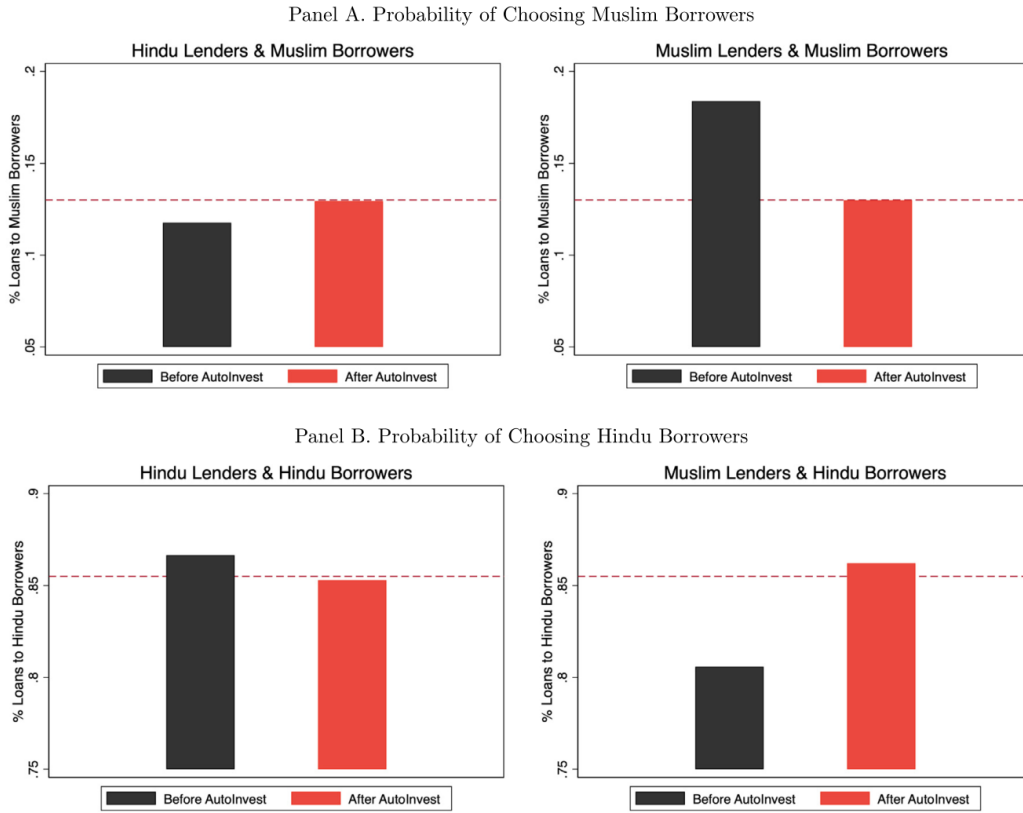


Fig. 1. Raw before/after religious lending shares.

Read.

- Before AUTO INVEST, Hindu lenders fund Muslim borrowers less often than the Muslim share in the platform pool, while Muslim lenders fund Muslim borrowers more often than the platform share.
- After AUTO INVEST, both groups move much closer to the platform-wide borrower composition.

Why it matters.

- This is the cleanest raw-data picture of in-group bias.

- The convergence after robo-advice already suggests that the tool is removing demographic favoritism rather than simply changing overall platform composition.

Table 2: Main regression: change in lending to out-group borrowers

Dependent variable: Muslim borrower					Low use Auto invest	High use Auto invest
	(1)	(2)	(3)	(4)	(5)	(6)
Hindu Lender $\times$ Auto Invest	0.045** (2.51)	0.046** (2.51)	0.045** (2.07)	0.043** (1.96)	0.009 (0.23)	0.052* (1.94)
Hindu Lender	-0.058*** (-3.52)	-0.058*** (-3.54)				
Auto Invest	-0.026 (-1.45)	-0.025 (-1.40)	-0.030 (-1.43)	-0.033 (-1.53)	0.011 (0.29)	-0.048* (-1.83)
Maturity		0.012*** (4.78)	0.009*** (3.20)	0.010*** (3.35)	0.010** (2.31)	0.009** (2.31)
Loan Amount (₹000)		-0.001*** (-10.34)	-0.001*** (-10.81)	-0.001*** (-10.92)	-0.001*** (-5.47)	-0.001*** (-9.27)
Interest Rate		-0.009 (-0.47)	-0.015 (-0.74)	-0.012 (-0.62)	0.013 (0.47)	-0.035 (-1.25)
Constant	0.181*** (11.14)	0.163*** (8.74)	0.119*** (11.72)	0.120*** (11.77)	0.099*** (6.77)	0.137*** (9.60)
Lender Fixed Effects			✓	✓	✓	✓
Year Fixed Effects				✓	✓	✓
N. obs.	113,284	113,283	113,283	113,283	39,366	72,104

This table reports the results of estimating the following specification by ordinary least squares:

$$\text{Muslim Borrower}_{i,j,t} = \alpha + \beta \text{Auto Invest}_{j,t} + \gamma \text{Hindu Lender}_j + \delta \text{Hindu Lender}_j \times \text{Auto Invest}_{j,t} + \zeta \mathbf{x}_{i,t} + \eta_j + \eta_t + \epsilon_{i,j,t},$$

where $\text{Muslim Borrower}_{i,j,t}$ is equal to 1 if the borrower i who receives funding from lender j in year t is Muslim, and zero otherwise; $\text{Auto Invest}_{j,t}$ is equal to 1 if the lender made the loans after activating Auto Invest and 0 otherwise; Hindu Lender_j is equal to 1 if lender j is Hindu; and $\mathbf{x}_{i,t}$ is a vector of loan-level characteristics that are direct proxies for the risk profiles of the loans lenders extend to borrowers—loan

Table 2 Multivariate evidence on lending to Muslim borrowers.

Read.

- In the simplest specification, the coefficient on **Hindu lender** is about -0.058 : before AUTO INVEST, Hindu lenders are about **5.8 percentage points less likely** than Muslim lenders to fund Muslim borrowers.
- The interaction **Hindu lender $\times$ Auto Invest** is about $+0.045$ to $+0.046$, which means robo-advice raises Hindu lenders' propensity to fund Muslim borrowers by about **4.5 percentage points**.
- The estimate is stable after adding loan characteristics, lender fixed effects, and year fixed effects.
- The effect is concentrated among high-intensity users of AUTO INVEST; low-intensity users show little change.

Interpretation.

- The main religion result is not a fragile raw correlation.
- It survives controls and fixed effects, which is exactly what we want in a before/after lender-level design.

Fig. 4. Heterogeneity by salience of Hindu-Muslim conflict

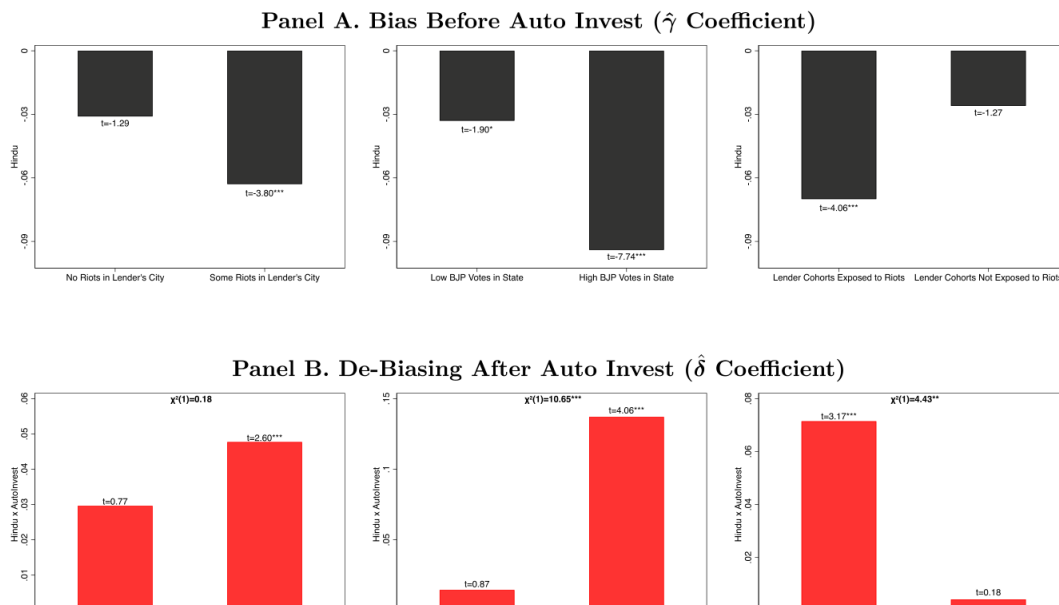


Fig. 4 Stronger pre-bias and stronger debiasing where inter-religious conflict is more salient.

Read.

- Pre-robo bias is much larger in places with more Hindu-Muslim riots, in states with higher BJP vote shares, and for younger lenders more exposed to modern Hindu-Muslim conflict during formative years.
- In the riot split, pre-bias is about **6.4 pp** for lenders in high-riot states and close to zero for others.
- In the BJP split, pre-bias is about **9.4 pp** in high-BJP states versus about **3.5 pp** elsewhere.
- Debiasing after adoption is also stronger where conflict is more salient.

Why it matters.

- This is powerful supporting evidence that the paper is picking up *cultural bias*, not a generic portfolio-rebalancing effect.

Fig. 5. Raw caste-based discrimination against Shudra borrowers

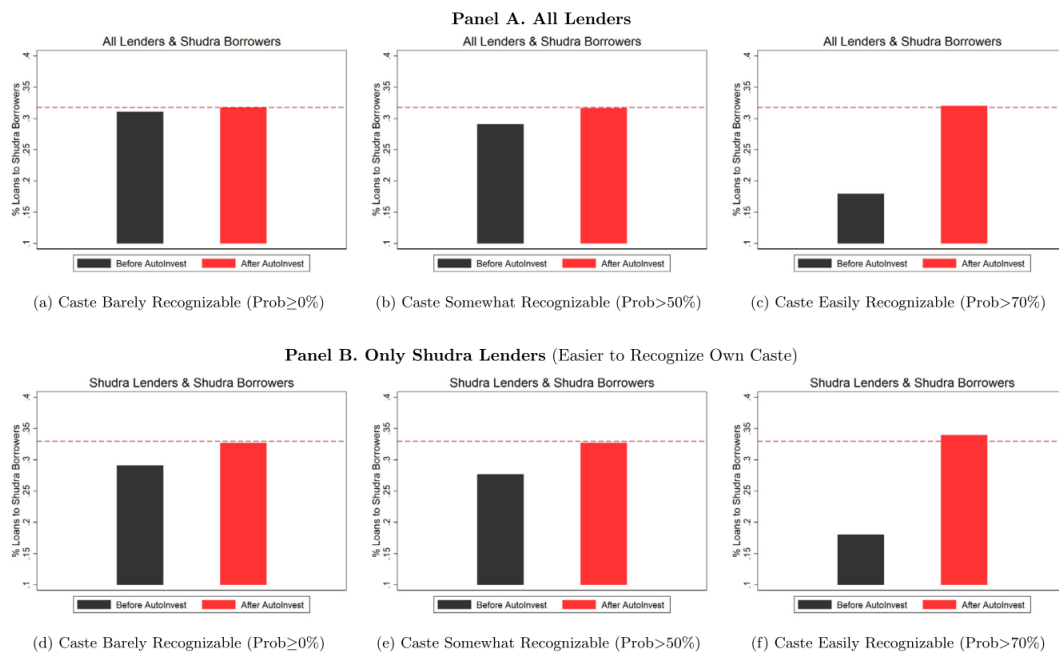


Fig. 5. Before/after lending to Shudra borrowers.

Read.

- The share of Shudra borrowers in lenders' portfolios rises after AUTO INVEST.
- The increase is especially strong when borrowers' caste is easier to recognize from observable characteristics.
- The same pattern appears, if anything more sharply, among Shudra lenders themselves.

Key regression counterpart (not reproduced separately here).

- The baseline estimate for lending to Shudra borrowers after adoption is about +0.011.
- Once caste recognizability exceeds 70% or 80%, the effect more than doubles.
- In states with more crimes against lower castes, the increase in lending to highly recognizable Shudra borrowers is around **2.7 pp**; elsewhere it is much smaller and often insignificant.

Interpretation.

- This is *stereotypical* discrimination, not just in-group favoritism: the platform-wide shift is against low-caste borrowers, and the bias is stronger when caste is easier to infer.

Table 3 & Fig. 8. Defaults: discriminated borrowers are safer ex ante

Dependent variable: Loan-level delinquency	In-group vs. out-group discrimination			Stereotypical discrimination		
	Hindu lenders			All lenders		
	(1)	(2)	Falsification (3)	(4)	(5)	Falsification (6)
Auto Invest	-0.108*** (-5.08)			-0.157*** (-6.25)		
Hindu Borrower× Auto Invest		-0.112*** (-5.21)	-0.072*** (-4.55)			
Muslim Borrower× Auto Invest		-0.073** (-2.49)	-0.070*** (-2.82)			
Muslim Borrower	-0.024** (-2.02)	-0.046** (-2.58)	-0.002 (-0.10)			
Non-Shudra Borrower× Auto Invest					-0.160*** (-6.04)	-0.100*** (-4.39)
Shudra Borrower× Auto Invest					-0.148*** (-4.54)	-0.104*** (-3.32)
Shudra Borrower				-0.038*** (-3.03)	-0.041*** (-2.89)	-0.030** (-2.14)
Constant	0.499*** (30.95)	0.501*** (30.84)	0.403*** (17.69)	0.490*** (32.13)	0.491*** (32.25)	0.421*** (7.77)
Loan Characteristics			✓			✓
Lender Fixed Effects			✓			✓
Year Fixed Effects			✓			✓
N. obs.	16,985	16,985	16,985	6821	6821	6821
R-Square	0.012	0.012	0.263	0.020	0.020	0.357

Table 3: Loan defaults before and after robo-advising.

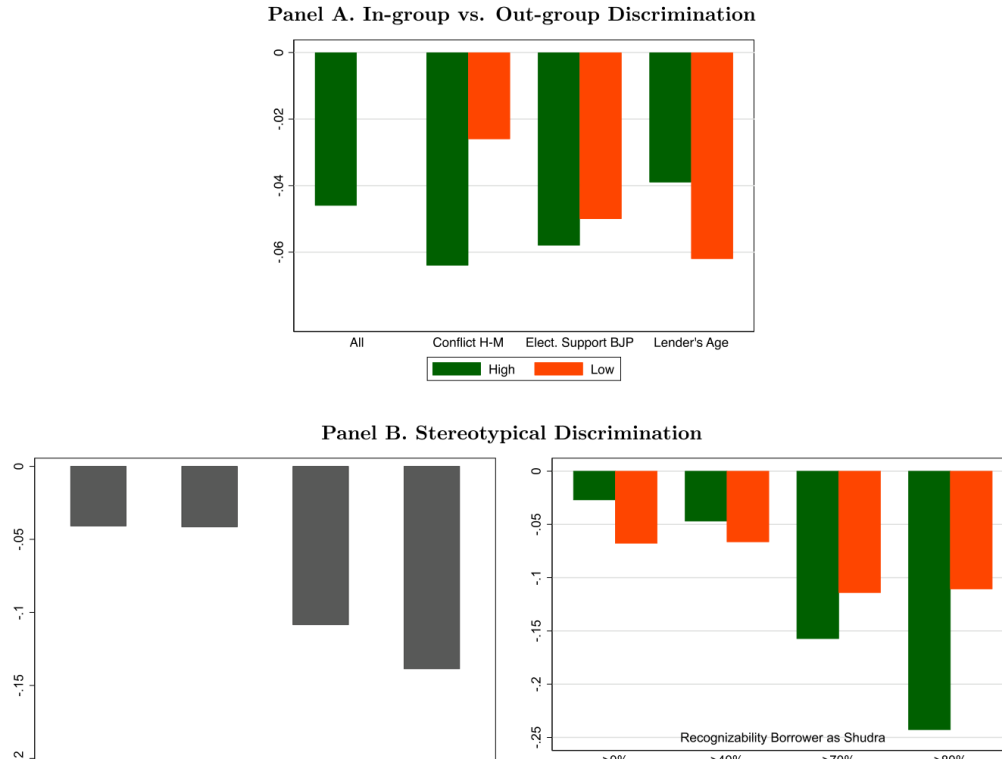


Fig. 8. Lower default of discriminated borrowers before robo-advice.

Read.

- In Hindu lenders' portfolios, **Muslim borrowers default less** than Hindu borrowers before robo-advice. In the default-regression table, the coefficient on **Muslim borrower** is about -0.046 in column (2).
- The companion default-gap figure shows the same result graphically: before AUTO INVEST, Muslim borrowers are roughly **4.6 percentage points less likely** to default than Hindu borrowers, and the gap is larger in conflict-salient environments.
- After adoption, defaults fall for both groups, but they fall *more* for favored-group loans. In the default-regression table, the **Hindu borrower** $\times$ **Auto Invest** term is more negative than the **Muslim borrower** $\times$ **Auto Invest** term.

Interpretation.

- Manual bias makes lenders choose *worse Hindu borrowers*, not better ones.
- This is the opposite of a benign "I know my own group better" story.
- The same basic logic shows up for caste: Shudra borrowers are less likely to default before robo-advice, and convergence happens after adoption.

Table 4 & Fig. 9. Repayment behavior and the non-diversification result

Dependent variable:	In-group vs. out-group discrimination			Stereotypical discrimination		
	St.dev loan Sizes (1)	St.dev loan Returns (2)	St.dev Payments (3)	St.dev loan Sizes (4)	St.dev loan Returns (5)	St.dev Payments (6)
Portfolio-level diversi- fication (<i>see columns</i>)						
Share Hindu Borrowers $\times$ Auto Invest	-0.004 (-1.11)	-0.023 (-0.31)	0.002 (1.39)			
Share Muslim Borrowers $\times$ Auto Invest	-0.003 (-0.18)	-0.301 (-0.72)	-0.006 (-0.82)			
Share Muslim Borrowers	0.014 (0.67)	-0.351 (-0.96)	0.005 (0.80)			
Share Non-Shudra Borrowers $\times$ Auto Invest				-0.006 (-0.84)	-0.215 (-1.51)	-0.002 (-1.01)
Share Shudra Borrowers $\times$ Auto Invest				-0.022** (-2.09)	0.178 (0.82)	0.003 (0.94)
Share Shudra Borrowers				0.012 (0.89)	-0.773*** (-3.81)	-0.002 (-0.64)
Constant	0.045*** (17.04)	0.632*** (10.77)	0.040*** (36.95)	0.038*** (7.55)	0.926*** (8.22)	0.043*** (26.14)
N	2586	399	2707	998	292	979
R-Square	0.001	0.018	0.001	0.014	0.033	0.001

This table reports the results of estimating variations of the following specification by ordinary least squares at the lender's level:

$$\begin{aligned} \text{Diversification Portfolio}_{ij} = & \alpha + \gamma \text{ Share Muslim Borrowers}_{ij} \\ & + \delta \text{ Share Muslim Borrowers}_{ij} \times \text{Auto Invest}_{ij} \\ & + \theta \text{ Share Hindu Borrowers}_{ij} \times \text{Auto Invest}_{ij} + \epsilon_{ij}, \end{aligned}$$

where $\text{Diversification Portfolio}_{ij}$ is computed at the lender level j before and after Auto Invest (i) in three ways: the standard deviation of the sizes of individual loans in lender j 's portfolio before and after Auto Invest (columns (1) and (4)), the standard deviation of the returns of the loans originated in lender i 's portfolio before and after Auto Invest (columns (2) and (5)), and the standard deviation of the size of the

Table 4: Lender portfolio diversification before and after robo-advising.

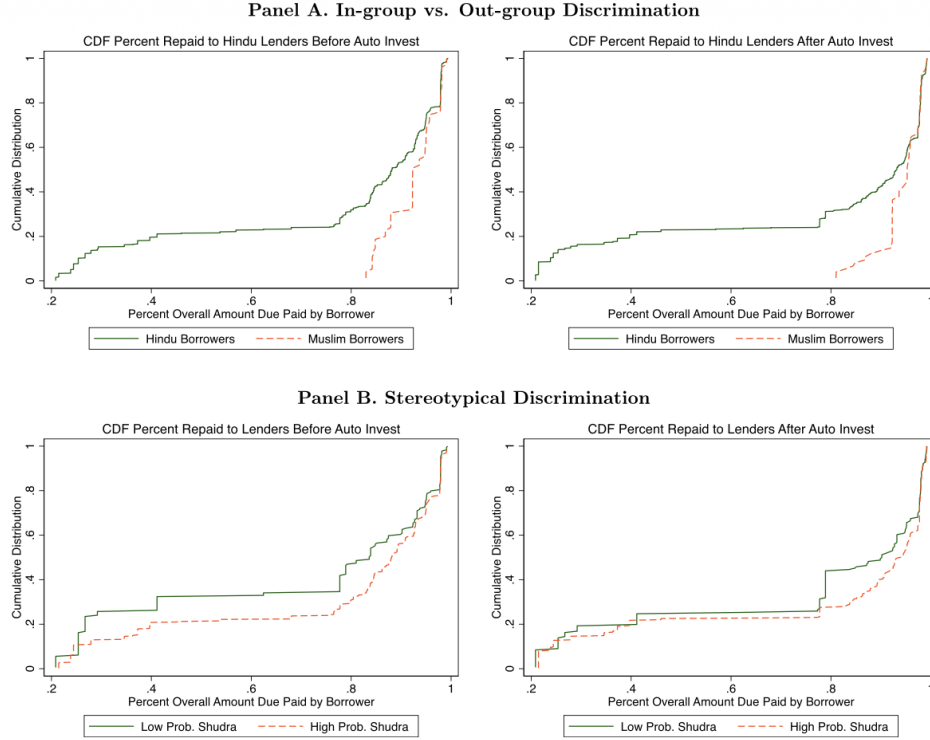


Fig. 9. Fraction of loan repaid before and after robo-advising. This figure plots a set of cumulative distribution functions (CDFs) for different groups of borrowers over the share of the overall amounts due (principal plus interest) that borrowers repaid by the time their loan account was closed. In all Panels, the left graph refers to the CDFs of borrowers in lenders' portfolios before moving to the robo-advising tool (Auto Invest). In contrast, the right graph refers to the CDFs after moving to Auto Invest. Panel A includes borrowers in Hindu lenders' portfolios. The solid green lines are the CDFs for Hindu borrowers, and the orange dashed lines are for Muslim borrowers. Panel B only includes Shudra borrowers in Hindu lenders' portfolios. The solid green lines are the CDFs for Shudra borrowers whose probability of caste recognition is below 15% based on the algorithm developed by [Bhagavatula et al. \(2018\)](#), and the orange dashed lines for other Shudra borrowers. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Fig. 9. CDFs of repayment shares before and after robo-advice.

Read the diversification table.

- Across portfolio-level dispersion in loan sizes, returns, and monthly payments, there is **no systematic diversification improvement** after AUTO INVEST.
- Most coefficients are small and insignificant. There is no coherent pattern strong enough to explain the return gains.

Read the repayment-distribution figure.

- Before robo-advice, the repayment distribution for favored-group borrowers is worse — especially in the left tail.
- After robo-advice, the repayment distributions converge.

Interpretation.

- The performance gains are not mainly about better diversification.
- They are about *which borrowers* lenders stop selecting — especially the very weak loans inside the favored demographic group.

Table 5: Returns: discriminated borrowers earn higher returns before robo-advice

Dependent variable: Loan-level return	In-group vs. out-group discrimination			Stereotypical discrimination		
	Hindu lenders			All lenders		
	(1)	(2)	Falsifi- cation (3)	(4)	(5)	Falsifi- cation (6)
Auto Invest	0.201*** (3.02)			0.300*** (2.14)		
Hindu Borrower× Auto Invest		0.222*** (3.07)	0.098 (0.78)			
Muslim Borrower× Auto Invest		−0.012 (−0.18)	−0.028 (−0.19)			
Muslim Borrower	0.282*** (6.25)	0.438** (5.13)	0.313*** (4.44)			
Non-Shudra Borrower× Auto Invest					0.307*** (2.98)	0.245 (1.12)
Shudra Borrower× Auto Invest					0.283** (2.33)	0.192 (0.79)
Shudra Borrower				0.106* (1.77)	0.122 (1.49)	0.257** (2.40)
Constant	0.361*** (5.09)	0.349*** (4.68)	−0.061 (−0.37)	0.314*** (3.05)	0.310*** (2.80)	−1.261*** (−5.51)
Loan Risk (Int.Rate)			✓			✓
Portfolio Risk (St.dev>Returns)			✓			✓
Lender Fixed Effects			✓			✓
Year Fixed Effects			✓			✓
N. obs.	2134	2134	2134	859	859	859

Table 5: Loan returns before and after robo-advising.

Read.

- In the religion sample, the coefficient on **Muslim borrower** is about **0.282** standard deviations in column (1): before robo-advice, Muslim borrowers generate *higher returns* than Hindu borrowers in Hindu lenders' portfolios.
- The paper interprets this as about **12.5 percentage points higher returns**.
- The coefficient on **Auto Invest** is about **0.201** standard deviations, which means average returns rise after adoption.
- The return improvement is driven almost entirely by the loans to favored Hindu borrowers: **Hindu borrower × Auto Invest** is about **0.222**, whereas the corresponding Muslim interaction is near zero.
- With loan-risk and portfolio-risk controls included, these differences shrink substantially or disappear.

Interpretation.

- The core mistake is not under-investing in Muslims *per se*; it is over-investing in low-quality favored Hindu borrowers when choosing manually.
- For caste, the same pattern appears: Shudra borrowers earn higher returns before robo-advice, and the post-adoption gains mostly come from the previously favored non-Shudra side.

Table 6: Quantile regressions: the gains come from removing the left tail

Panel A. In-group vs. out-group discrimination								
Dependent variable:	OLS		25th percentile		Median		75th percentile	
	Hindu (1)	Muslim (2)	Hindu (3)	Muslim (4)	Hindu (5)	Muslim (6)	Hindu (7)	Muslim (8)
Loan's Return								
Auto-Invest (Without Loan characteristics)	0.217*** (4.92)	0.065** (2.25)	0.250* (1.82)	0.173** (2.23)	0.274*** (11.22)	0.121*** (3.84)	0.109*** (11.66)	-0.004 (-0.29)
Auto-Invest (With Loan characteristics)	-0.160*** (-6.15)	-0.020* (-1.66)	-0.028 (-0.91)	-0.010 (-0.69)	-0.026 (-0.90)	-0.006 (-0.41)	-0.008 (-0.79)	0.000 (0.00)
N. obs.	2326	220	2326	220	2326	220	2326	220
Panel B. Stereotypical Discrimination								
Dependent variable:	OLS		25th percentile		Median		75th percentile	
	Low Prob Shudra (1)	High Prob Shudra (2)	Low Prob Shudra (3)	High Prob Shudra (4)	Low Prob Shudra (5)	High Prob Shudra (6)	Low Prob Shudra (7)	High Prob Shudra (8)
Auto-Invest (Without Loan characteristics)	0.298*** (2.89)	0.259*** (4.54)	1.838*** (4.61)	0.304* (1.73)	0.332*** (3.02)	0.272*** (8.13)	0.114*** (5.64)	0.094*** (9.29)
Auto-Invest (With Loan characteristics)	-0.097* (-1.72)	-0.145*** (-4.12)	-0.016 (-0.15)	-0.013*** (-2.06)	-0.000 (-0.00)	-0.001 (-0.13)	-0.006 (-0.42)	-0.003 (-1.55)
N. obs.	462	1158	462	1158	462	1158	462	1158

This table reports the results of estimating the following set of quantile regressions:

$$Q_\tau(\text{Loan's Return}_{i,j}) = \alpha(\tau) + \beta(\tau) \text{Auto Invest}_j + X'_{i,j} \zeta(\tau) + \epsilon_{i,j},$$

whose outcome variable is quantile Q_τ of the distribution of the (standardized) loan return associated with borrower i and lender j throughout the sample period; Auto Invest_j is

Table 6: Quantile regressions for loan returns.

Read.

- Without risk controls, AUTO INVEST improves returns most strongly in the **25th percentile** and **median** of the return distribution.
- The improvement fades as one moves toward the 75th percentile.
- Once loan-risk controls are included, the quantile effects largely vanish.

Interpretation.

- Robo-advice is not generating a broad uniform boost to all loans.
- It is mainly eliminating the *worst* manual loans — the bad left-tail favored-group loans that biased lenders selected on their own.

Fig. 10. & Fig. 11. How much debiasing, and how much money?

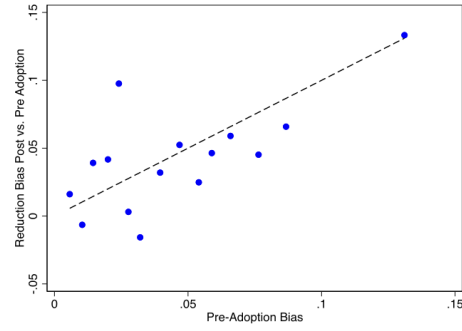


Fig. 10. Lender-level drop in ethnic bias after robo-advising relative to before. This figure is a binned scatterplot of the reduction in ethnic bias computed at the individual lender level against the level of bias before adoption. For each lender, the reduction is computed as the difference between the extent of bias in the loans issued by the lender before robo-advice adoption and the extent of bias after adoption. A positive value indicates that the bias before adoption is higher than the bias after adoption. The dashed line is a 45-degree line, which represents the hypothetical case in which the pre-adoption bias and the reduction in bias are identical. That is, the bias after adoption is zero.

Fig. 10. Lender-level drop in ethnic bias versus pre-adoption bias.

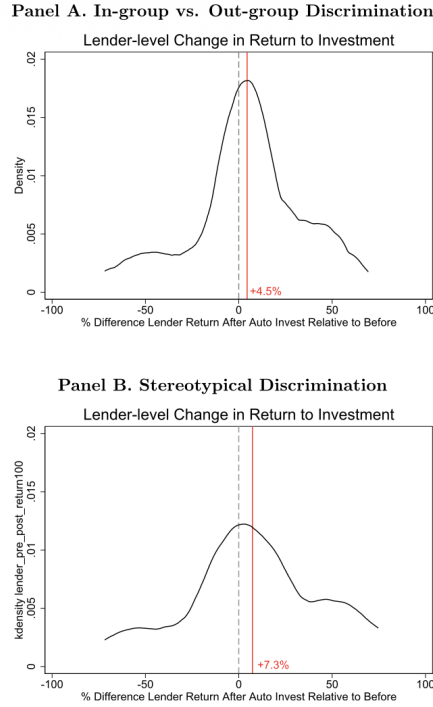


Fig. 11. Lender-level change in returns after robo-advising. This figure plots the density of the lender-level average change in the returns to loans originated on the platform after using Auto Invest relative to before. To compute the change, we first compute the overall return to the amounts invested by each lender separately before and after using Auto Invest across all loans they originated. We then subtract the two returns and average these changes across lenders. Loan-level returns are thus value-weighted within lenders, and lender-level returns are equally weighted across lenders. Panel A refers to Hindu lenders' loans to Hindu borrowers (in-group vs. out-group discrimination), whereas Panel B refers to all Hindu lenders' loans to Shudra borrowers (stereotypical discrimination).

Fig. 11. Lender-level change in returns after robo-advising.

Read the bias-change figure.

- Lenders with more pre-adoption bias tend to experience a larger drop in bias after adoption.
- Many lenders debias substantially, but a large share do not debias *fully* — consistent with the fact that even adopters can still allocate some funds manually.

Read the return-change figure.

- Average lender-level total return rises by about **4.5 percentage points** in the religion sample.
- Average lender-level total return rises by about **7.3 percentage points** in the caste sample.

Interpretation.

- These are economically meaningful gains from debiasing.
- The paper further shows that almost all of the gain comes from the returns on previously *favored* groups, which is exactly what the over-selection story predicts.

6 Short summary

- **Method.** Compare the same lenders' manual P2P lending choices with their choices after adopting a demographically blind-ish robo-advisor on a platform that already screens and prices borrowers.
- **Conceptual result.** Cultural bias in finance can hurt the biased decision-maker, not just the discriminated-against party.
- **Main empirical result.** Manual lenders favor co-ethnic and socially favored borrowers; robo-advice reduces that favoritism.
- **Mechanism result.** Biased lenders choose *worse-quality favored-group borrowers*. The improvement from robo-advice comes from eliminating those low-quality loans, especially in the left tail of returns.
- **Quantitative result.** Unassisted lending is associated with roughly 8% higher defaults and up to 7.3 percentage-point lower returns; lender-level returns rise by about 4.5 pp to 7.3 pp after adoption.
- **Policy lesson.** A simple FinTech tool that removes demographic sorting from the last-stage matching process can generate meaningful debiasing and performance gains.

7 Conclusion

- **Core conclusion.** The paper presents unusually clean field evidence that cultural bias affects real-world risk-taking and portfolio allocation, and that those biases can be privately costly.
- **Economic intuition.** When left to themselves, lenders tilt toward socially favored groups and end up selecting lower-quality borrowers within those groups. AUTO INVEST does not make lenders smarter in a deep structural sense; it prevents them from expressing the demographic tilt at the point of borrower matching.
- **Why the interpretation is compelling.** The signs of the performance results, the salience heterogeneity, the lack of diversification effects, and the rarity of overrides all point toward biased beliefs rather than efficient discrimination or a pure taste for discrimination.
- **Broader takeaway.** FinTech can be a force for efficiency not only because it lowers transaction costs, but also because it can suppress predictable human distortions in choice under risk.
- **Caveat.** External validity should still be handled carefully: this is a setting with strong platform screening, non-negotiated pricing, and little relationship lending. But *within* that environment, the paper makes a strong case that cultural debiasing improves both fairness and returns.